

BCT-SMOKE

OHC Resonance-Guided Selective Pyrolysis and Void-Geometry Filter Architecture

Michel Robert Cabrié · Independent Artist & Researcher

Barrys Reef, Victoria, Australia · ORCID: 0009-0007-9561-9859

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CLAIMS

1. (independent)

A method of reducing carcinogen formation during combustion of a combustible plant material, comprising: applying electromagnetic radiation at one or more frequencies derived from OHC Josephson resonance multiples of BCT void geometry parameters $r_{oct} = (\sqrt{2}-1)/2$ and $r_{tet} = (\sqrt{6}-2)/4$ to the combustion zone of said plant material during combustion, wherein said frequencies are selected to couple to polycyclic aromatic hydrocarbon (PAH) ring-closure transition states, thereby suppressing PAH formation.

2. (dependent on 1)

The method of claim 1, wherein the applied frequency is in the range 10-100 THz, corresponding to OHC resonance multiples of r_{oct} in the mid-infrared C=C aromatic stretch region.

3. (dependent on 1)

The method of claim 1, wherein the applied frequency additionally couples to N-nitroso bond formation transition states at a frequency derived from ω_J/r_{tet} , thereby suppressing tobacco-specific nitrosamine (TSNA) formation.

4. (dependent on 1)

The method of claim 1, wherein the combustible plant material is tobacco and the applied frequency is selected to preserve nicotine transmission at greater than 90% of unmodified combustion levels.

5. (dependent on 1)

The method of claim 1, wherein the combustible plant material is cannabis and the applied frequency is selected to preserve cannabinoid transmission including THC and CBD at greater than 90% of unmodified combustion levels.

6. (independent)

A filter for a smoking device comprising an adsorption substrate having a pore geometry characterised by a mean pore radius of $R_{pore} = R_{nicotine} \cdot (1 + r_{oct})$, where $r_{oct} = (\sqrt{2}-1)/2$ is the BCT octahedral void radius and $R_{nicotine}$ is the effective kinetic radius of

nicotine, said pore geometry providing selective retention of polycyclic aromatic hydrocarbon molecules relative to nicotine vapour transmission.

7. (dependent on 6)

The filter of claim 6, wherein the adsorption substrate comprises a zeolite material with pore diameter in the range 3.8-4.6 Å.

8. (dependent on 6)

The filter of claim 6, wherein the adsorption substrate comprises a metal-organic framework (MOF) with D4-symmetric node geometry providing tunable pore size at BCT void geometry length scales.

9. (dependent on 6)

The filter of claim 6, wherein the selectivity ratio of PAH retention to nicotine transmission is maximised at pore geometry satisfying $R_{\text{pore}}(\text{PAH})/R_{\text{pore}}(\text{nicotine}) = r_{\text{oct}}/r_{\text{tet}} = (\sqrt{2}-1)/(\sqrt{6}-2) \times 2 \approx 1.843$.

10. (dependent on 6)

The filter of claim 6, wherein the filter additionally provides selective retention of tobacco-specific nitrosamines while permitting cannabinoid transmission including THC and CBD.

11. (independent)

A smoking device or adapter comprising in combination: the electromagnetic emitter of claim 1, operating at OHC resonance frequencies to suppress PAH formation during combustion; and the selective filter of claim 6, providing downstream capture of residual PAH and TSNA compounds; wherein combined operation reduces total carcinogen delivery by greater than 80% while maintaining greater than 90% nicotine or cannabinoid transmission.

12. (dependent on 11)

The smoking device of claim 11, wherein the electromagnetic emitter is powered by thermoelectric harvesting from the combustion zone.

13. (dependent on 11)

The smoking device of claim 11, configured for use with tobacco.

14. (dependent on 11)

The smoking device of claim 11, configured for use with medicinal cannabis.

15. (independent)

Use of BCT void geometry parameters $r_{\text{oct}} = (\sqrt{2}-1)/2$ and $r_{\text{tet}} = (\sqrt{6}-2)/4$ to determine: (a) electromagnetic frequencies for selective disruption of PAH ring-closure during pyrolysis; and/or (b) pore geometry dimensions for selective adsorption filter materials; in the manufacture of a harm-reduction device for combustible plant material including tobacco and

cannabis.

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Michel Robert Cabrié · ORCID 0009-0007-9561-9859 · ZeroFreeParameters@gmail.com · BCT Programme v47k · June
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